RF-Lens IQ Signal Characterization Report

Source: examples\data\synthetic_qpsk.sigmf-data

Generated: 2026-10-26T23:34:37+00:00

RF-Lens version: 0.1.0

Configuration hash: 70c8adc132fc

Objective

Characterize the recorded IQ spectrum using the configured spectral analysis pipeline.

Data and Methodology

Sample rate: 1e+06 Hz

Center frequency: 4.3392e+08 Hz

Samples: 82,011

Duration: 0.082011 s

Source metadata: {"datatype": "cf32_le", "sigmf_global": {"core:datatype": "cf32_le",

"core:description": "RF-Lens synthetic QPSK; seed=20261027; inband_snr_db=10",

"core:sample_rate": 1000000.0, "core:version": "1.0.0"}}

Processing: two-sided Welch PSD with configured window, segment length, and overlap.

Window: hann

Resolution bandwidth (ENBW): 732.422 Hz

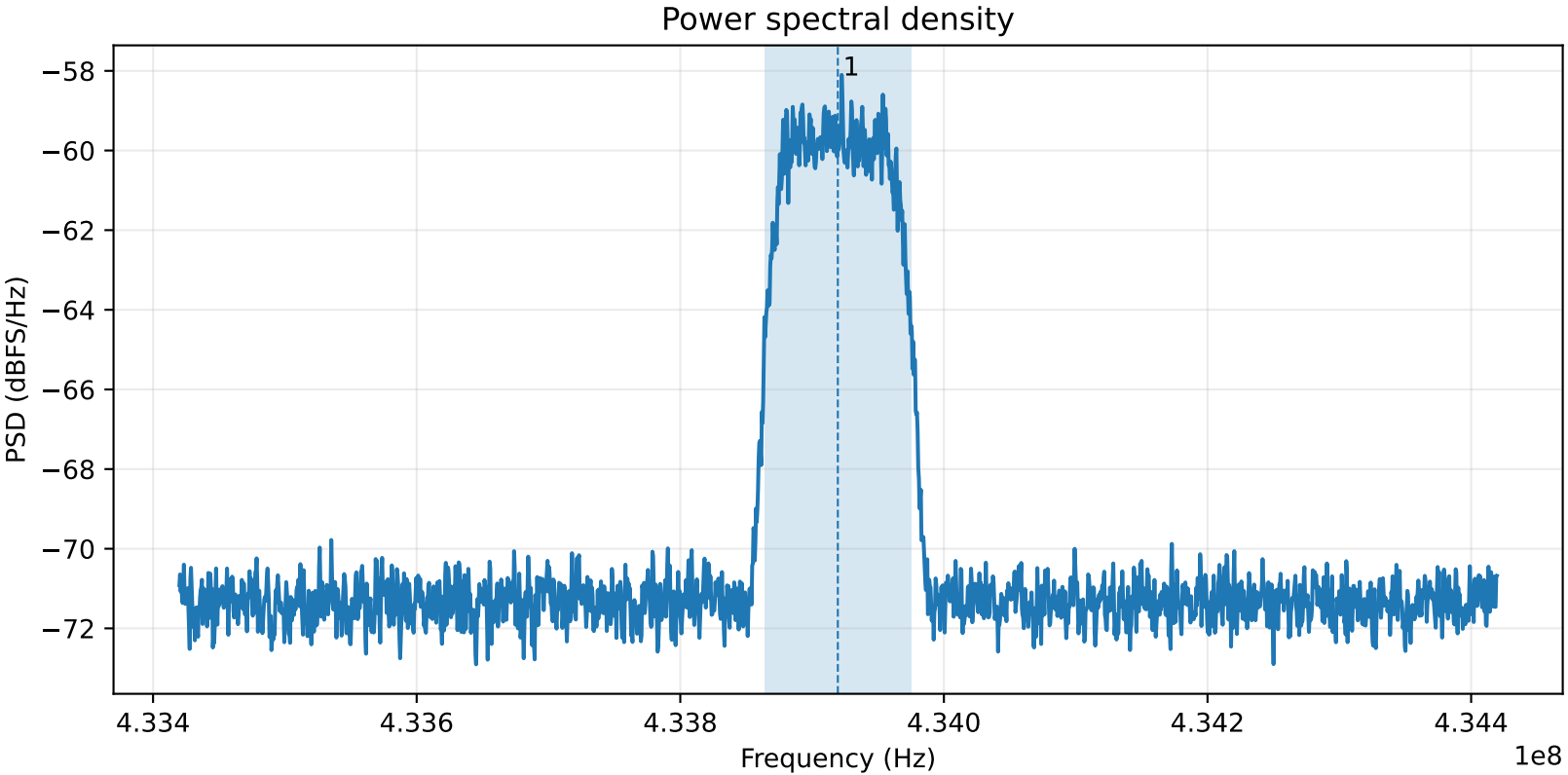
Welch segments: 79

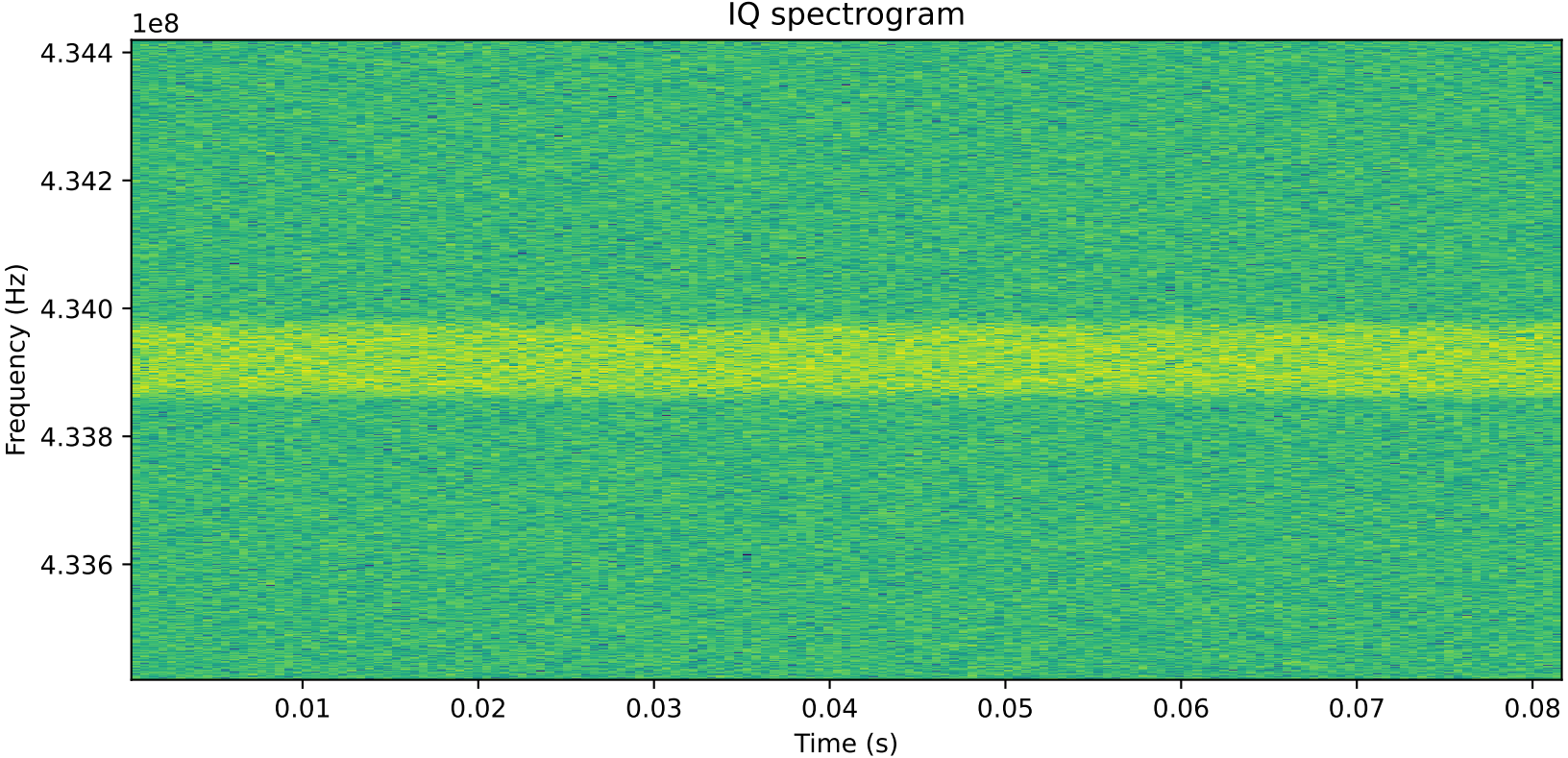
Noise density: -71.309 dBFS/Hz

OBW99 uses 0.5% tail exclusion on each side after noise subtraction.

SNR is in-band signal power divided by estimated noise over the measured bandwidth.

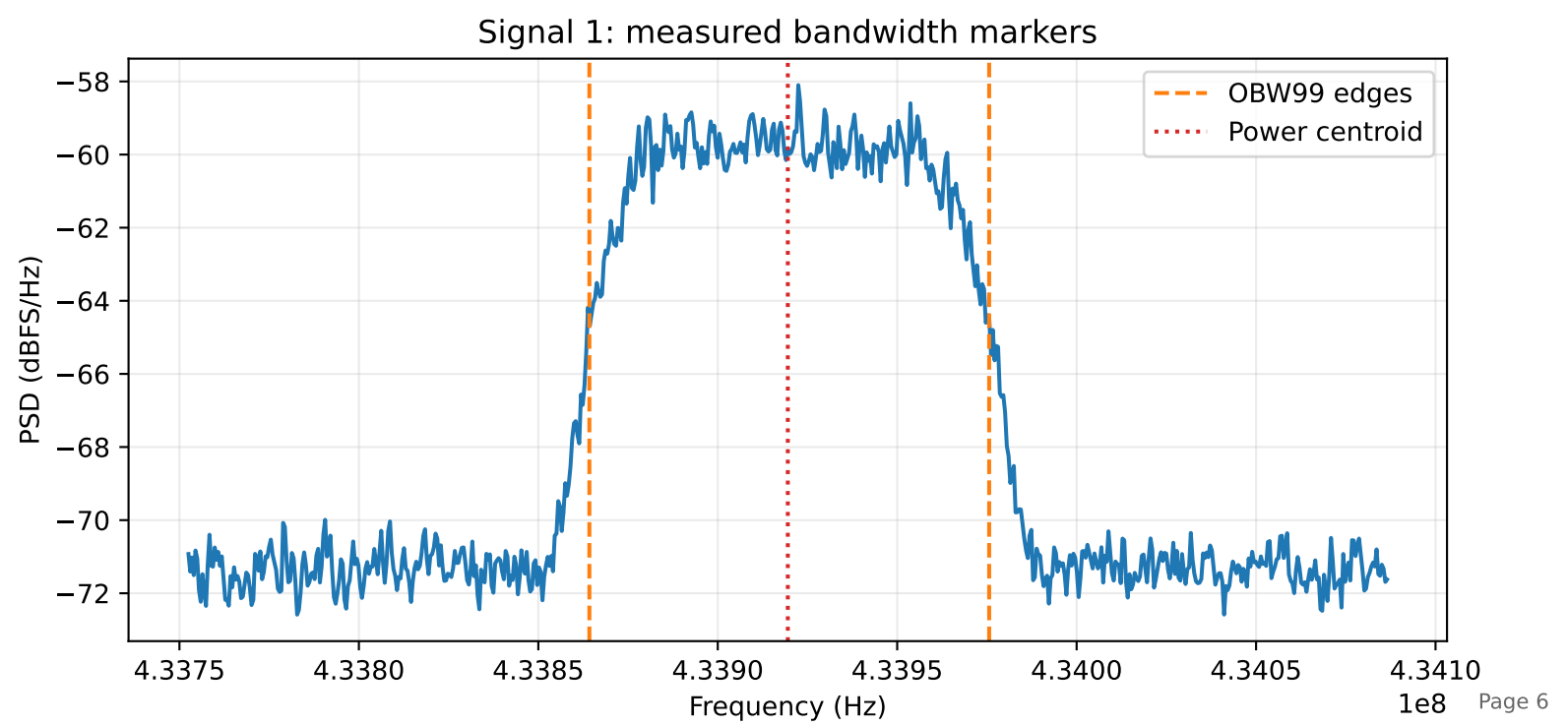
Absolute calibrated power in dBm is not available in v1.


Power spectral density
PSD (dBFS/Hz)
−58
−60
−62
−64
−66
−68
−70
−72
1
4.334
4.336
4.338
4.340
4.342
4.344
Frequency (Hz)
1e8


IQ spectrogram
1e8
4.344
4.342
4.340
4.338
4.336
Frequency (Hz)
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Time (s)

Signal Measurements

ID	Center (Hz)	OBW99 (Hz)	-3 dB (Hz)	-26 dB (Hz)	Power (dBFS)	SNR (dB)	Flags
1	433919522.108	111396.603	93466.165	unavailable	-10.05	10.60	bw_26db_unavailable


Signal 1: measured bandwidth markers
OBW99 edges
Power centroid
PSD (dBFS/Hz)
−58
−60
−62
−64
−66
−68
−70
−72
4.3375
4.3380
4.3385
4.3390
4.3395
4.3400
4.3405
4.3410
1e8
Frequency (Hz)

Signal 1: observations

Measured center: 433919522.108 Hz

OBW99: 111396.603 Hz

Power: -10.053 dBFS

In-band SNR: 10.604 dB

Flags: bw_26db_unavailable

These observations are generated from the measurement values above.

Limitations and Conclusions

Limitations

No absolute power calibration is applied. Frequency and bandwidth precision are limited by the PSD bin spacing and selected analysis settings.

Noise-floor and measurement estimates depend on the signal and noise assumptions documented in the methodology.

Conclusions

Detected signals: 1

Warnings: none

The conclusions on this page are restricted to the measurements and warnings reported in this analysis.

Appendix: Configuration and Software

Python: 3.14.3

numpy: 2.3.4

scipy: 1.16.2

matplotlib: 3.10.7

rflens: 0.1.0

Analysis configuration:

{"merge_gap_hz": 0.0, "min_width_bins": 3, "noverlap": 1024, "nperseg": 2048, "threshold_db": 6.0, "window": "hann"}

Formulas: PSD power conversion uses 10 log10(P). OBW99 retains 99% of integrated noise-subtracted power.

Signal power integrates positive noise-subtracted PSD across the selected measurement region.

SNR is signal power divided by estimated noise density integrated over that region.